

36.5 Summary and Notes

36.5.1 Summary

Idea.

1. The multinomial distribution is an exponential family with natural parameters the baseline-category logits, cumulant $b(\boldsymbol{\eta}) = \log(1 + \sum_s e^{\eta_s})$, mean $m\nabla b$ and covariance $m(\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top)$ (Lemma 36.1.1). The chapter then chooses which $c - 1$ logits to make linear.
2. The baseline-category logit model (Definition 36.1.1) is the canonical-link generalized linear model for that family: score $\mathbf{X}^\top(\mathbf{y}_r - \boldsymbol{\mu}_r)$ per category, observed information equal to expected, and a log-likelihood strictly concave when $\text{rank}(\mathbf{X}) = p$, so the maximum is unique when it exists (Theorem 36.1.2).
3. Changing the baseline is a linear reparameterization, leaving estimates, fitted values, the maximized likelihood and the standard errors of contrasts unchanged (Proposition 36.1.3). A coefficient is a log odds ratio against the baseline, not a derivative of a probability (Proposition 36.1.4).
4. For an ordered response, modelling $P(Y \leq r)$ uses the ordering (Definition 36.2.1). Fitted probabilities are positive only where the linear predictors stay in order: automatic with common slopes, a real restriction without them (Proposition 36.2.1). Score and expected information are $\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1}(\tilde{\mathbf{y}}_i - m_i \tilde{\boldsymbol{\pi}}_i)$ and $\sum_i m_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i$ (Proposition 36.2.2), and the log-likelihood is concave when the link density is log-concave.
5. The proportional odds model has a latent-variable derivation, slopes that survive any collapsing of adjacent categories, cutpoints that never need constraining, and one odds ratio applying at every cut (Theorem 36.3.1). The collapsing invariance is also how it is checked.
6. That restriction is tested by a score statistic computed from the restricted fit alone, on $(c - 2)|A|$ degrees of freedom, and diagnosed by plotting the slopes of the $c - 1$ dichotomized fits (Proposition 36.3.2). When it fails: partial proportional odds, a different link, a scale effect, a different family, or an honest table of slopes.
7. A sequential model (Definition 36.4.1) models the probability of stopping at each stage. Its likelihood factors into $c - 1$ binomial likelihoods, so with stage-specific slopes it is $c - 1$ separate binary regressions with asymptotically independent estimates (Theorem 36.4.1), and with a common slope one binary regression on expanded data (Corollary 36.4.2). No ordering constraint is needed.
8. With the complementary log-log link the two ordinal families coincide, cutpoints related by a cumulative sum on the exponential scale (Proposition 36.4.3): the grouped-data proportional hazards model. With the logit link they differ, and the data can tell them apart.

36.5.2 Notes and sources

Coverage. The topics follow Agresti (2015, chapter 6) and Fahrmeir, Kneib, Lang and Marx (2021, §6.2–6.3); the treatment, proofs, examples and exercises are our own.

Baseline-category logits. The model grew up independently in biostatistics and econometrics.

McFadden (1974) derived it from random utility maximization, which is the source of the name *multinomial logit* and of the discrete-choice literature; the underlying axiom is Luce's (1959), and the independence of irrelevant alternatives in Exercise (C2) is its most famous consequence and problem. McFadden's conditional logit model, in which regressors vary by category, is the standard extension, not treated here. Existence of the maximum likelihood estimate is settled by Albert and Anderson (1984); Theorem 35.3.1 is the binary case. The individualized fits of the remark in [Section 36.1](#) are analysed by Begg and Gray (1984).

Cumulative models. Modelling cumulative probabilities with a common slope goes back to Walker and Duncan (1967). McCullagh (1980) named the proportional odds model and gave the latent-variable derivation, the general cumulative link family and the invariance properties of Theorem 36.3.1; McCullagh and Nelder (1989, chapter 5) is a compressed version. Anderson (1984) introduced a different ordinal family, the *stereotype* model, which estimates category scores rather than assuming them. That the log-likelihood of a cumulative link model is concave whenever the link density is log-concave was proved by Pratt (1981) and, for grouped data specifically, by Burrige (1981); it is the reason Fisher scoring is as docile here as it is. The asymptotics of [Section 36.2](#) are imported rather than proved, this model not being a generalized linear model; the standard accounts are van der Vaart (1998) and Lehmann and Casella (1998). Agresti (2010) is the book-length treatment, and covers the scale-effect models of [Section 36.3](#) and the connection with rank statistics in Exercise (C2).

Checking proportional odds. The score test of Proposition 36.3.2 is the one most software reports; Brant (1990) gave the Wald version based on the separate dichotomized fits, which is where the habit of plotting them comes from; it tests the same hypothesis as the score statistic but is a different statistic. Partial proportional odds models are due to Peterson and Harrell (1990), who also noted that such fits can produce negative probabilities — the failure of Proposition 36.2.1.

Sequential models. Continuation-ratio logits go back to Fienberg (1980); the sequential formulation used here, with a link and a linear predictor at each stage, is Tutz's (1991). Fahrmeir and Tutz (2001) and Tutz (2012) give the textbook treatments, the latter with a taxonomy of ordinal models built out of binary ones. The factorization of Theorem 36.4.1 is the algebra that underlies discrete-time survival analysis, where the stages are time intervals and the continuation ratios are hazards; the continuous-time ancestor is Cox (1972), and Proposition 36.4.3, due to Läärä and Matthews (1985), connects the two.

Data. Both examples use public-domain data shipped with statsmodels: the travel-mode data of Example (Choosing how to travel) come from the online archive of Greene's econometrics text, and Example (A seven-point scale) uses an extract of the 1996 American National Election Study, whose education and income variables are ordered scales treated here as numerical scores. Treating them as factors changes the coefficients but no conclusion about proportional odds. Neither example is a causal statement, for the reasons in [Chapter 25](#).

What is missing. Responses clustered, repeated or overdispersed relative to Lemma 36.1.1 need more than this chapter offers: [Chapter 38](#) takes up quasi-likelihood and overdispersion, [Chapter 40](#) random effects and estimating equations, and [Chapter 39](#) Bayesian fitting, where the latent variable of Theorem 36.3.1(a) becomes a data augmentation scheme.

36.5.3 References

- Agresti, Alan (2010). *Analysis of Ordinal Categorical Data*. 2nd edition. Hoboken, NJ: Wiley.
- Agresti, Alan (2015). *Foundations of Linear and Generalized Linear Models*. Hoboken,

NJ: Wiley.

- Albert, Adelin and Anderson, John A. (1984). On the Existence of Maximum Likelihood Estimates in Logistic Regression Models. *Biometrika* 71(1), 1–10.
- Anderson, John A. (1984). Regression and Ordered Categorical Variables. *Journal of the Royal Statistical Society, Series B* 46(1), 1–30.
- Begg, Colin B. and Gray, Robert (1984). Calculation of Polychotomous Logistic Regression Parameters Using Individualized Regressions. *Biometrika* 71(1), 11–18.
- Brant, Rollin (1990). Assessing Proportionality in the Proportional Odds Model for Ordinal Logistic Regression. *Biometrics* 46(4), 1171–1178.
- Burridge, J. (1981). A Note on Maximum Likelihood Estimation for Regression Models Using Grouped Data. *Journal of the Royal Statistical Society, Series B* 43(1), 41–45.
- Cox, David R. (1972). Regression Models and Life-Tables. *Journal of the Royal Statistical Society, Series B* 34(2), 187–220.
- Fahrmeir, Ludwig and Tutz, Gerhard (2001). *Multivariate Statistical Modelling Based on Generalized Linear Models*. 2nd edition. New York: Springer.
- Fahrmeir, Ludwig, Kneib, Thomas, Lang, Stefan and Marx, Brian D. (2021). *Regression: Models, Methods and Applications*. 2nd edition. Berlin: Springer.
- Fienberg, Stephen E. (1980). *The Analysis of Cross-Classified Categorical Data*. 2nd edition. Cambridge, MA: MIT Press.
- Läärä, Esa and Matthews, John N. S. (1985). The Equivalence of Two Models for Ordinal Data. *Biometrika* 72(1), 206–207.
- Lehmann, Erich L. and Casella, George (1998). *Theory of Point Estimation*. 2nd edition. New York: Springer.
- Luce, R. Duncan (1959). *Individual Choice Behavior: A Theoretical Analysis*. New York: Wiley.
- McCullagh, Peter (1980). Regression Models for Ordinal Data. *Journal of the Royal Statistical Society, Series B* 42(2), 109–142.
- McCullagh, Peter and Nelder, John A. (1989). *Generalized Linear Models*. 2nd edition. London: Chapman and Hall.
- McFadden, Daniel (1974). Conditional Logit Analysis of Qualitative Choice Behavior. In Paul Zarembka (editor), *Frontiers in Econometrics*, 105–142. New York: Academic Press.
- Peterson, Bercedis and Harrell, Frank E. (1990). Partial Proportional Odds Models for Ordinal Response Variables. *Journal of the Royal Statistical Society, Series C* 39(2), 205–217.
- Pratt, John W. (1981). Concavity of the Log Likelihood. *Journal of the American Statistical Association* 76(373), 103–106.
- Tutz, Gerhard (1991). Sequential Models in Categorical Regression. *Computational Statistics and Data Analysis* 11(3), 275–295.
- Tutz, Gerhard (2012). *Regression for Categorical Data*. Cambridge: Cambridge University Press.
- Walker, Strother H. and Duncan, David B. (1967). Estimation of the Probability of an Event as a Function of Several Independent Variables. *Biometrika* 54(1/2), 167–179.
- van der Vaart, Aad W. (1998). *Asymptotic Statistics*. Cambridge: Cambridge University Press.